

Data Visualization

Data visualization is the graphical representation of data using charts, graphs, and plots. It helps in understanding patterns, trends, and insights in a clear and effective manner.

Python provides several libraries for data visualization such as Matplotlib, Seaborn, Plotly, and Bokeh.

Matplotlib

Matplotlib is a popular Python library used for creating static, animated, and interactive visualizations. It was introduced by John Hunter in 2002 and is built on NumPy.

Features of Matplotlib

- Easy to use
- Highly customizable
- Supports many plot types

Graph types in Matplotlib

1. Line Plot:

Description: A line plot connects data points with a continuous line.

Use Case: Used to show trends over time

```
import matplotlib.pyplot as plt

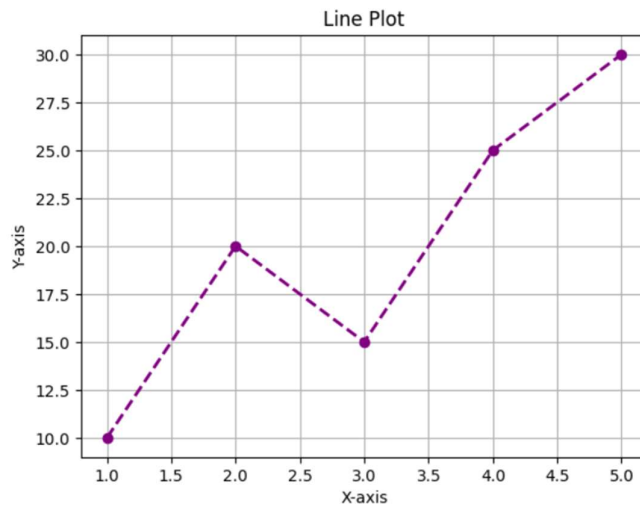
x = [1, 2, 3, 4, 5]
y = [10, 20, 15, 25, 30]

plt.plot(x, y, color='purple', marker='o', linestyle='--', linewidth=2)

plt.title("Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.grid(True)

plt.show()
```

Output:



2. Bar Chart:

Description: Represents data using rectangular bars.

Use Case: Used for comparing categories.

```
import matplotlib.pyplot as plt

categories = ['A', 'B', 'C', 'D']
values = [10, 25, 15, 20]

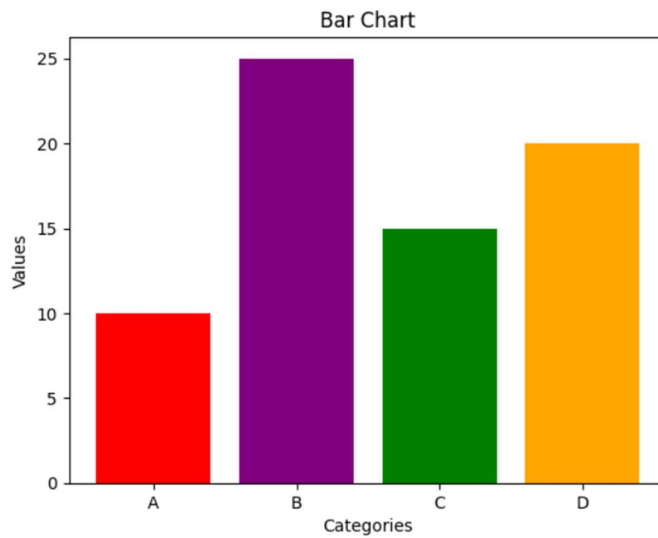
colors = ['red', 'purple', 'green', 'orange']

plt.bar(categories, values, color=colors)

plt.title("Bar Chart")
plt.xlabel("Categories")
plt.ylabel("Values")

plt.show()
```

Output:



3. Histogram:

Description: Displays data distribution using bins.

Use Case: Used to understand frequency distribution.

```
import matplotlib.pyplot as plt

data = [5, 7, 8, 7, 2, 4, 6, 7, 8, 9, 5, 3, 4, 6, 7]

plt.figure(figsize=(8,5))

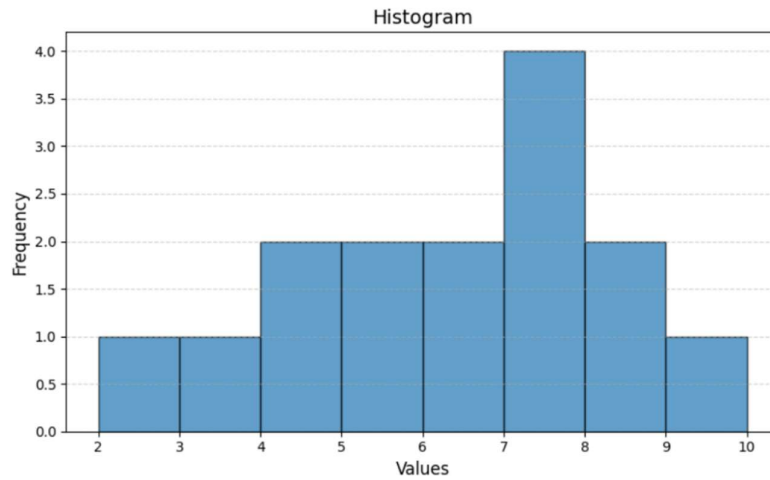
plt.hist(data, bins=range(2,11),
         edgecolor='black',
         alpha=0.7)

plt.title("Histogram", fontsize=14)
plt.xlabel("Values", fontsize=12)
plt.ylabel("Frequency", fontsize=12)

plt.grid(axis='y', linestyle='--', alpha=0.5)

plt.tight_layout()
plt.show()
```

Output:



4. Scatter Plot:

Description: Uses dots to represent data points.

Use Case: Shows relationship between variables.

```
import matplotlib.pyplot as plt
import numpy as np

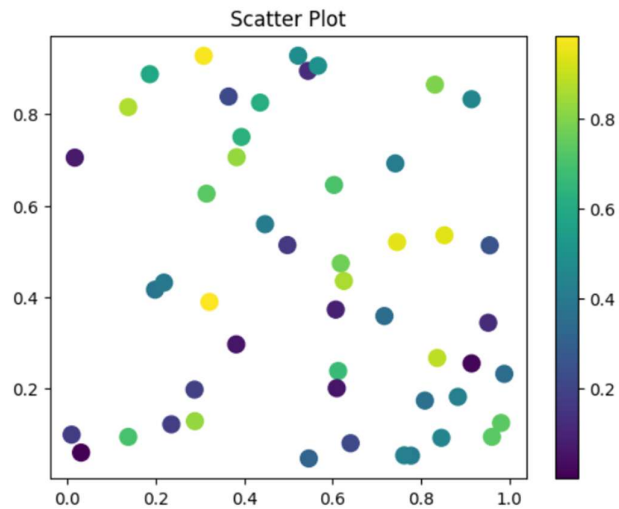
x = np.random.rand(50)
y = np.random.rand(50)
colors = np.random.rand(50)

plt.scatter(x, y, c=colors, cmap='viridis', s=100)

plt.title("Scatter Plot")
plt.colorbar()

plt.show()
```

Output:



5. Pie Chart:

Description: Circular chart divided into sectors.

Use Case: Shows percentage distribution.

```
import matplotlib.pyplot as plt

sizes = [30, 20, 25, 25]
labels = ['A', 'B', 'C', 'D']
colors = ['red', 'yellow', 'green', 'blue']

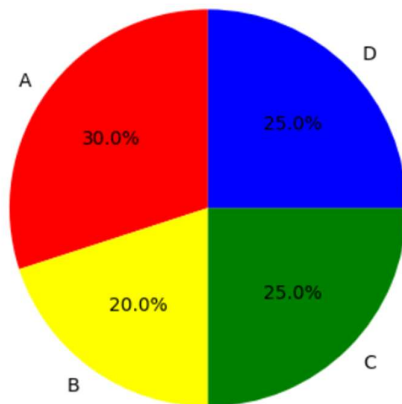
plt.pie(sizes, labels=labels, colors=colors,
        autopct='%1.1f%%', startangle=90)

plt.title("Pie Chart")

plt.show()
```

Output:

Pie Chart



Seaborn

Seaborn is a high-level data visualization library built on top of Matplotlib. It provides better design, attractive colour palettes, and simpler syntax for complex visualizations.

Features

- Built-in themes and styles
- Easy-to-use functions
- Better statistical visualization

Graph types in Seaborn

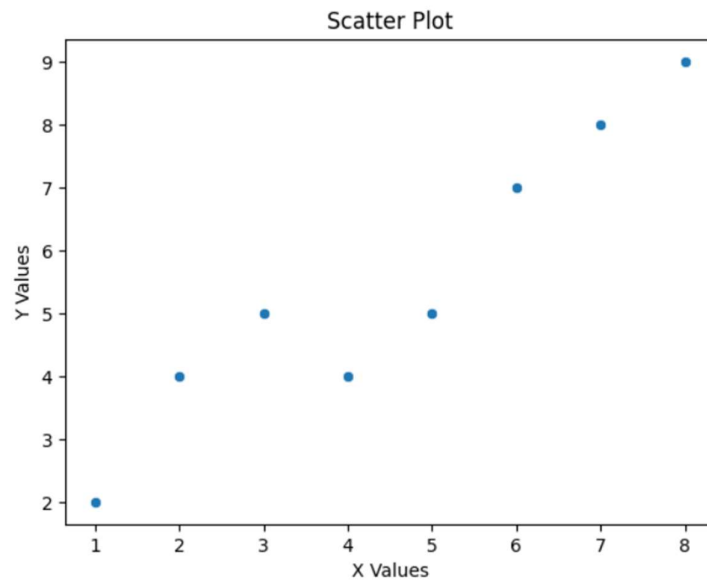
1. Scatter plot:

Description: Displays relationship between variables.

Use Case: Used for correlation analysis.

```
import seaborn as sns
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5, 6, 7, 8]
y = [2, 4, 5, 4, 5, 7, 8, 9]
sns.scatterplot(x=x, y=y)
plt.title("Scatter Plot")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.show()
```

Output:



2. Histogram:

Description: Shows distribution of data.

Use Case: Used for frequency analysis

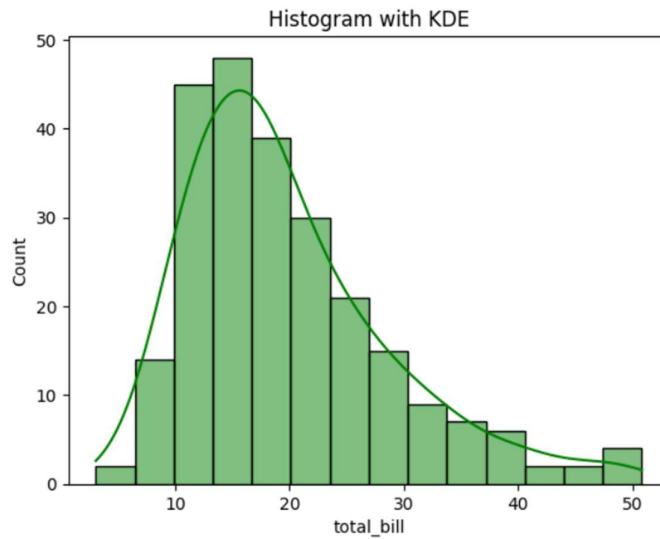
```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

sns.histplot(data=tips,
              x="total_bill",
              kde=True,
              color="green")

plt.title("Histogram with KDE")
plt.show()
```

Output:



3. Box Plot:

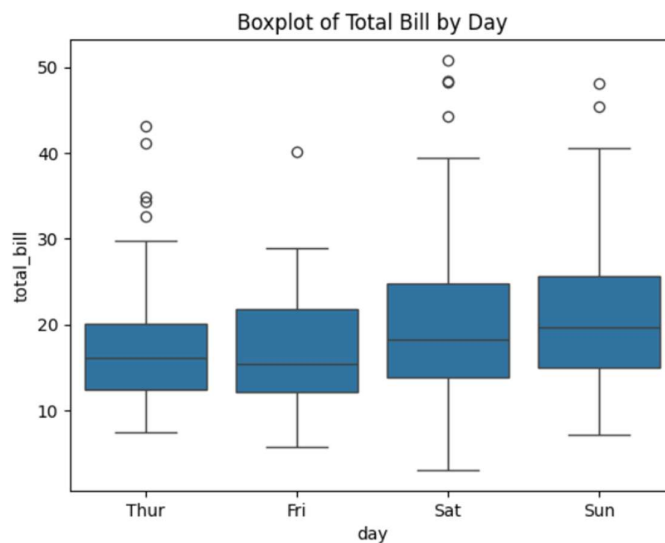
Description: Displays distribution using quartiles.

Use Case: Used to detect outliers.

```
import seaborn as sns
import matplotlib.pyplot as plt

data = sns.load_dataset("tips")
sns.boxplot(x="day", y="total_bill", data=data)
plt.title("Boxplot of Total Bill by Day")
plt.show()
```

Output:



4. Heatmap:

Description: Uses colours to represent data values.

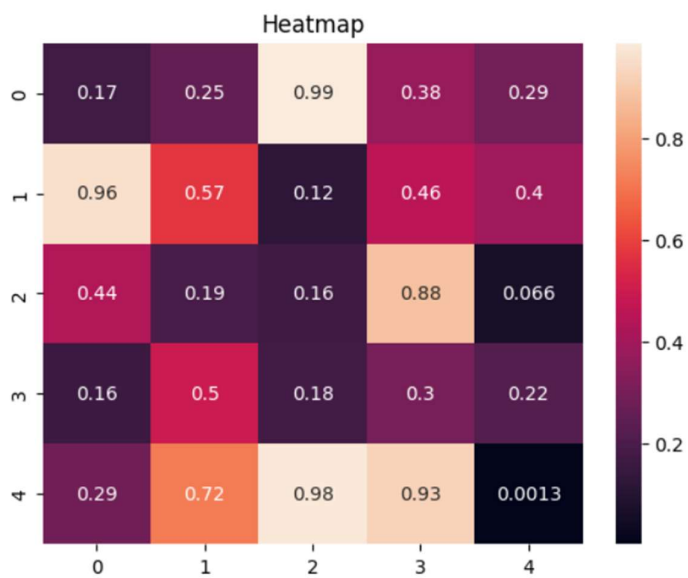
Use Case: Used for correlation matrices.

```
import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt

data = np.random.rand(5, 5)

sns.heatmap(data, annot=True)
plt.title("Heatmap")
plt.show()
```

Output:



Differences between matplotlib and seaborn:

Mathplotlib vs Seaborn

Mathplotlib	Seaborn
Matplotlib is a low-level data visualization library that allows users to build plots from scratch with full control.	Seaborn is a high-level visualization library built on top of Matplotlib that simplifies plotting and improves design.
Matplotlib is comparatively harder to use because it requires more lines of code and manual effort.	Seaborn is easier to use because it provides simpler functions and requires less code.

The default appearance of Matplotlib plots is basic and plain, and users need to customize them manually.	Seaborn plots are visually appealing by default, with built-in themes and better color combinations.
Matplotlib mainly works with basic data types such as lists and arrays.	Seaborn works very well with Pandas DataFrames and can directly use column names.
Matplotlib is mainly used for basic plots like line graphs, bar charts, histograms, and scatter plots.	Seaborn is used for both basic and advanced statistical plots like box plots, violin plots, heatmaps, and pair plots.
Matplotlib offers a high level of customization, allowing detailed control over every part of the graph.	Seaborn offers limited customization compared to Matplotlib but focuses on simplicity and aesthetics.
Matplotlib is best suited for creating highly customized and complex visualizations.	Seaborn is best suited for quick, attractive, and statistical data visualizations.